KARAN DATTA DAPHADE

Data & AI Engineer | Product Builder | Founder, eatmit
Jalna, Maharashtra (open to relocate) | karandaphade001@gmail.com | LinkedIn
Live apps: QField Dashboard • GDD Application • Farmer AI Chatbot

PROFILE

I build and run the data and software that field teams actually use. As the only tech/data intern for Materra's Jalna–Wardha cluster for 7 months, I took field survey data from raw mobile exports to cleaned datasets, automated pipelines, AI-enabled tools and deployed dashboards, working directly with agronomists and field executives. Before that I spent years in product design (Figma, user flows, user stories), and I am now building eatmit, a restaurant POS platform. I learn new tools quickly and put them to work straight away, and I care most about simple tools that work for people on the ground.

EXPERIENCE

Data Intern, Digital Agronomy – Materra India Pvt. Ltd. *Jul 2025 – Feb 2026*
Jalna, Maharashtra. Sole data/tech resource for the Jalna–Wardha cluster; worked with agronomy, ML and field teams.

- **Field data pipeline:** automated the daily merge, cleaning and standardisation of QField survey data in Python (Pandas), replacing 4–5 hours of manual reporting each day across 140+ farms and 1,000+ field visits. Also exported clean GeoJSON/CSV for DSSAT and ML.
- **GDD crop model as a web app:** turned a CLI-only Growing Degree Days model into a multi-page Streamlit app for agronomists: daily Open-Meteo ingestion with gap detection into PostgreSQL, 11×11 km grid clustering to cut API calls, and predictions for 10 cotton phenophases. Used daily by agronomy and ML teams.
- **AI-enabled farmer advisory:** led a 15-day WhatsApp AI activation in Wardha and Jalna, onboarding 100+ farmers in 3 sessions; built an APMC mandi price scraper (agmarknet.gov.in, 36 states, 5,000+ records on active days) that fed daily prices into the advisory.
- **KML farm boundary QA tool (v0.1 to v0.4):** Python pipeline using UTM projection to detect overlapping farms and wrong area values, with colour-coded KML output for field corrections and Google Drive API I/O. Unblocked regenerative certification audits; improved over four versions from field feedback.
- **Dashboards and field support:** built and deployed a Streamlit dashboard so management could track each Field Executive's progress daily, and supported FEs on QField issues and data consistency.
- **Procurement workflow (Google AppSheet):** redesigned the cotton procurement workflow end to end: deal creation, gin arrival tracking, bale traceability and payments. Defined the business rules and user flows, then built a separate version for IDH procurement operations.
- **Field work:** collected IDH baseline survey data (7 days, Dhar, MP) and scripted the download of 1,400+ survey images for audit.

Freelance Product / UI-UX Designer *2022 – Present*

- **Product design:** designed 7+ web and mobile products (landing pages, dashboards, apps) for small businesses and startups: user flows, wireframes and prototypes in Figma, plus user stories, and tracking delivery timelines with clients.
- **Feedback and hand-off:** ran user feedback rounds on prototypes and revised flows before hand-off to developers; worked with small teams to keep scope, quality and timelines on track.
- **Founder, eatmit (restaurant POS SaaS, in development):** researched the product, defined use cases, user flows and business logic, and am building a POS and live-operations platform: table ordering, KDS, live order dashboard, billing and menu control by day and time.

SKILLS

Data engineering: Python (Pandas, NumPy), SQL, PostgreSQL, Supabase, ETL/ELT pipelines, REST APIs, web scraping, scheduled jobs (Windows Task Scheduler), Apache Airflow (basic)
Apps and dashboards: Streamlit (multi-page, deployed), Google AppSheet, QField, Excel, data reporting and monitoring
Tools: Git, GitHub, Linux (Ubuntu), Google Drive API, Claude Code, Antigravity, Bash (basic)
Product: Figma, Jira, user stories and requirement writing, user flows and wireframing, user testing, defining business logic
Geospatial: QGIS, KML/GeoJSON processing, UTM projection, farm boundary QA
Domain: Regenerative cotton farming, field survey operations, GDD crop modelling, agri commodity pricing (APMC)

EDUCATION

Bachelor of Computer Applications (BCA), Deogiri College, BAMU University, Chh. Sambhajinagar – 2024

LANGUAGES

English (fluent), Hindi (native), Marathi (native)